Review Article

Harnessing the Power of Combination of Herbal Extracts against UTI Pathogens

Amruta Prakashrao Kanakdande* | **Shubham Pathade**

School of Life Sciences, Swami Ramanand Teerth Marathwada University, Nanded, India

Use your device to scan and read the article online

Citation A.P. Kanakdande, S. Pathade, **Harnessing the Power of Combination of Herbal Extracts against UTI Pathogens.** *Eurasian J. Sci. Technol.*, **2026**, 6(3), 287-293.

https://doi.org/10.48309/ejst.2026.570634.1301

CC BY

Article info:
Received: 2026-01-08
Accepted: : 2026-06-29
Available Online: 2026-07-15
ID: EJST-2601-1301

Keywords:
Amla; Antibiotics; Brahmi; Herbal extracts; Urinary-tract infections

ABSTRACT

Antibacterial efficacy through a synergistic approach of selective herbal extracts, *i.e.,* amla (*Emblica officinalis*) druhalad (*Berberis aristata*) and brahmi (*Bacopa monnieri*) is evaluated to discover an alternative approach to control urinary-tract infections (UTIs) caused by common uropathogenic bacteria. These herbs are extracted using a Soxhlet extraction method in presence of methanol and water as solvents. The separate extracts of all medicinal herbs in both the solvents are evaluated for their antibacterial potential against common uropathogens, isolated from UTI patients. Based on morphological and biochemical tests, two microorganisms were identified as *i.e., Bacillus subtilis* and *Staphylococcus aureus.* On the basis of 16S rRNA sequencing techniques, the one was identified as *Kosakonia cowanii (*OQ 073698*)*, alongwith the yeast *Candida albicans*. Overall, significant zones of inhibition are recorded against all the uropathogens considered under study upon amla:daruhalad 7:3 (w/w) ratio in water extract with extracted yield 9.0 g/250 mL. One of the core findings from the present investigation includes the uropathogen *Kosakonia cowanii* is resistant to ampicillin while the above-selected screened combination of extracts is an effective and alternative treatment approach for ampicillin resistant uropathogens. This synergistic effect achieved with the tested extract combinations can provide a starting point for the development of alternative formulations, effective against the antibiotic resistant *Kosakonia cowanii* (OQ073698) isolate which would be effective against other uropathogens. Additionally, it was proven that these edible plant extracts could be used as natural alternative inhibitors to control microbial pathogenesis related to UTIs.

Introduction

Nowadays, urinary tract infections (UTIs) are the most common human infections across the globe. Indeed, it has been estimated that nearly 800 million people (equating to approximately 11% of the global population) face UTI at least once in a year [1,2]. However, recurrent UTIs (rUTIs) are more frequent among young females, with approximately 50% of women having experienced a UTI at least once in their lives

*Corresponding Author: bioamruta@gmail.com (A.P. Kanakdande)

[3,4]. An appropriate choice and dose of antibiotics have been cured these infections in many patients, but sometimes the infection occurs more frequently in about 25% of the infected patients [5]. Uropathogenic microorganisms are basically from Gram-negative and Gram-positive classes; certain fungi and sometimes yeast can cause the UTIs [2]. The common uropathogenic *Escherichia coli* (UPEC) covers the highest rate of infections *i.e.,* 51-75%, while others include *Klebsiella, Enterococcus, Staphylococcus* and *Candida* species as well[6-10]. Because the majority of bacteria cause UTIs and antibiotics are the general standard of care but, sometimes frequent use of antibiotics may lead to develop resistance in the bacteria. To solve this issue, there is an urgent need to explore new sources of antimicrobials. According to the World Health Organization (WHO), medicinal herbs will be a valuable asset for upcoming era of new medication therapy [11]. The herbal remedy is under development in health care and demands more attention [12].

Generally, plants are considered as a valuable source of traditional treatment since ancient times [13]. The WHO estimates approximately 80% of the world's population uses at least one traditional remedy by extracting active components from the plants (WHO, [14]). This is due to medicinal flora, offering various advantages over modern drugs, as modern drugs are expensive and often associated with hazardous side effects as well [15]. Herbal traditional medicines are more effective in overcoming the bacterial resistance developed by uropathogens. The herbal extracts offer high efficacy, easy availability, and minimal side effects [16].

This has prompted the researchersto explore new antimicrobial agents, particularly from traditional medicinal plants [17]. Considering the problem, the current study emphasizes on the isolation of UTI pathogens, extraction of herbs by using Soxhlet apparatus and antimicrobial activity against selectively isolated uropathogens.

Materials and Methods

Isolation of Uropathogens from UTI Patients

The urine samples of UTI patients were collected in aseptic bottles and were stored in refrigerator until next process. The urine samples were spread on UTI agar plates, separately. The plates were incubated in an incubator for 24-48 h. The selectively pure cultured colonies were picked and transferred them to UTI agar slants for further study. One of the colonies confirming the highest growth in UTI agar was further carried forward for 16S rRNA sequencing technique [18].

Extraction of Selective Herbal Plants

The Soxhlet extractions were performed with three plants *i.e.,* Aamla (*Emblica officinalis*), Brahmi (*Bacopa monnieri*), and Daruhalad (*Berberis aristata*) using methanol and water as solvents. Initially, the separate plant extractions were conducted and were extracted. The Soxhlet extraction was performed with a powder (25 g) of selected herbal plants in 250 mL of respective solvents. In second attempt, the Soxhlet extraction was performed with all permutations and combinations of the above-selected herbal extracts in 3:1, 4:2, 5:1, 6:1, and 7:3 proportions.

Antimicrobial Resistance Study

The uropathogens, isolated from UTI patients, were carried forward to test the antimicrobial resistance study. One of the bacteria, *i.e., Kosakonia cowanii* (OQ 073698) was tested for antimicrobial resistance patterns with the standard antibiotics considered under study such as; ampicillin (10 µg), streptomycin (10 µg), tetracycline (30 µg), gentamicin (120 µg), piperacillin (30 µg), and penicillin G (10 units).

Antimicrobial Activity of Herbal Extracts

The extracted yields in respective solvents of all the extracts considered under the study were calculated using following formula.

$$\text{Percentage Yield (\%)} = (\text{Dry weight of extract} / \text{Dry weight of plant material}) \times 100 \quad (1)$$

Furthermore, these extracts were used to evaluate the antimicrobial study. The antimicrobial study was performed using the agar well diffusion method. The sterile Luria Bertani agar plates were prepared. The test organisms *i.e.,* UTI pathogens were separately inoculated in nutrient broth and incubated overnight at 37 °C to adjust the turbidity 0.5 McFarland standards giving final inoculum concentration of 1.5×10^8 CFU/mL. A well of 6 mm was bored into sterile inoculated LB agar plates using a sterile cork borer (6 mm).

All the methanolic extracts were dissolved in DMSO prior to proceeding with the antimicrobial study, at a concentration of 1 mg/mL. However, the water as a solvent extracts were used as they were, with the same 1 mg/mL concentration. Thus, the as-prepared plates were incubated in incubator at 37 °C for 24 h to develop the zones of inhibition around the wells, which were recorded with the respective extracts against each uropathogen.

Phytochemical Tests of Selective Herbal Extracts

Based on the antimicrobial activity results, the selectively screened herbal extracts, were carried forward for the phytochemical analysis for flavonoids, alkaloids, saponins, tannins, etc.

Results and Discussion

Isolation of Uropathogen

Based on the biochemical tests and 16S rRNA sequencing analysis, the isolated uropathogen was identified as *Kosakonia cowanii* (OQ 073698) [18]. Other organisms isolated from UTI patients were carried forward for the biochemical identification, and based on biochemical tests, they were identified as *Bacillus subtilis* and *Staphylococcus aureus* (Table 1) [19,20].

Table 1: Morphological and biochemical test

Morphological and biochemical test Name	*Bacillus subtilis*	*Staphylococcus aureus*
Gram staining	Positive	*Positive*
Catalase	+	+
Citrate	+	
Flagella	Flagellated	Non flagellated
Gelatin hydrolysis	+	+
Urease	-	+
VP	+	+
Oxidase	variable	-
Nitrate reduction	+	+
MR	-	+
Indole	-	-
Motility	Motile	Non motile

All the biochemical tests showed positive results toward catalase, citrate, gelatin hydrolysis, nitrate reduction, and Vogues proskaeur, whereas negative for methyl red, gas production, indole, and urease tests. In case of *Staphylococcus aureus,* few tests were performed like catalse and VP was showed to be positive while oxidase showed negative. One of the traditional methods *i.e.,* selective media test was used for identification of *Staphylococcus aureus*. On Baird Parker agar medium, the *Staphylococcus aureus* colonies showed black and shiny colour with narrow white margins, surrounded by clear zone extending into the opaque medium [21]. However, the yeast *C.albicans* were not directly isolated from UTI patients, whereas it is under repository of SRTM University and directly used for the antimicrobial study.

Extraction of Herbal Medicines

Today, there are increasing numbers of UTIs with microbes that are multidrug resistant. As a

solution for this problem, there has been a renewed interest in alternative antimicrobials.

For the present investigation, based on the previous reports by the author Kanakdande *et al.,* three plants were selected. Herbal plant materials were collected from the local market and were scientifically authenticated from SLS, SRTMU Nanded. The *Emblica officinalis, Bacopa monnieri*, and *Berberis aristata* were screened for further study. The individual extract yield was reported to be the highest in 7:3 proportion of *Emblica officinalis + Berberis aristata* (A+D), followed by *Emblica officinalis+Bacopa monnieri* (A+B) and *Emblica officinalis* (A), *Bacopa monnieri* (B), and *Berberis aristata* (D). The ideal ratio for showing the highest extract yield in 7:3 ratio and were screened for further antimicrobial study. The highest extract yield was reported with the *Emblica officinalis + Berberis aristata* (A+D), in water, *i.e.*, 9 g/250 mL followed by methanol, *i.e.*, 7 g/250 mL (the data are not shown). The overall highest yield was reported with *Emblica officinalis* in water extract (Figure 1).

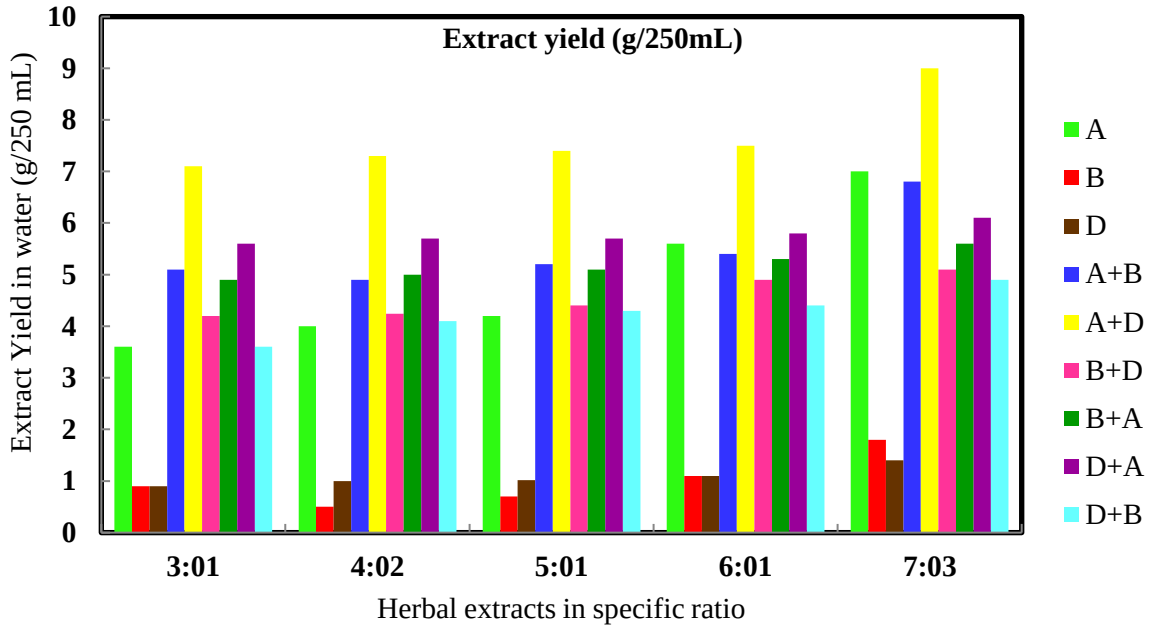


Figure 1: Extraction of herbal medicine from water using the Soxhlet extraction method

Antimicrobial Resistance

The selective uropathogens were tested for antimicrobial resistance patterns. Based the obtained results, it can be inferred that all isolated uropathogens were resistant to ampicillin, gentamicin, and penicillin antibiotics, while tetracycline was sensitive toward *C.albicans*. Streptomycin was reported to be sensitive toward *Kosakonia cowanii* (OQ 073698) and *C. albicans.* Increasing antibiotic resistance, fueled by increasing use of antibiotics in humans and animals, has occurred by a combination of resistance mechanisms. Resistance occurs due to successive mutations, emerging chromosomal mutations, mobile genetic elements (MGEs) or a combination of these factors. MGEs, such as plasmids and transposons, enable rapid spread of resistance between bacteria which might be the possible reason for developing resistance among the tested uropathogens under study. In present study, ampicillin, gentamicin, and penicillin were resistant to all uropathogens *(Bacillus subtilis, Staphylococcus aureus, and Kosakonia cowanii)* tested under study except that streptomycin and tetracycline were sensitive toward isolated uropathogen *i.e., Kosakonia cowanii* (OQ 073698). With the present invention, this problem can be solved using the combinations of herbal extracts to treat UTIs [22].

Based on the results obtained from antibiotic resistance study and extracted yield, the herbal extracts with 7:3 ratio of all the methanolic and aqueous extracts were prepared for antimicrobial studies.

Antimicrobial Study

The combination of *Emblica officinalis* (A) and *Berberis aristata* (D) with 7:3 methanolic extract ratio showed the highest zone of inhibition in all uropathogens, *i.e.,* Kosakonia *cowanii* (OQ 073698)(25 mm), *Bacillus subtilis* (19 mm), *Staphylococcus aureus* (18 mm), and yeast *C.albicans* (23 mm) (Figures 2 and 3). The proper concentration with the correct combination was optimized to achieve the highest zone of inhibition against UTI pathogens. In addition to this, all the uropathogens were previously resistant to the ampicillin antibiotics , concluding the fact that the combination of *Emblica officinalis + Berberis aristata* (A+D) in a 7:3 ratio of methanolic extract can replace the ampicillin antibiotics. Pure extracts of *Emblica officinalis* were effective against all uropathogens considered in the study but synergistic effect was observed when *Emblica officinalis* was combined with the *Berberis aristata.* Plant remedies are a natural option for the long-term treatment of recurrent UTIs, especially in a synergistic antibiotic approach, as they have antimicrobial effects, and can reduce different symptoms such as anti-inflammatory, antipyretic, and analgesic effects which further helps to decrease the adverse effects of antibiotics. The combination of these two herbal extracts, *i.e., Emblica officinalis* and *Berberis aristata* confirmed the presence of flavonoids, alkaloids, saponins, and tannins, which might interfere with the antimicrobial results against the uropathogens. Medicinal plants synthesize secondary metabolites in response to an infection (induced defense), which act as antibiotics against pathogenic microorganisms. The antibacterial compounds found in herbal medicines can prevent or treat infections in humans. The antibacterial activities of these bioactive compounds are primarily mediated by two basic mechanisms: chemically interfering with the synthesis or function of essential bacterial components and/or evading the known antibacterial resistance mechanisms through bacterial protein biosynthesis, bacterial cell-wall biosynthesis, bacterial cell membrane destruction; bacterial DNA replication and repair, and inhibition of the metabolic pathways [22,23].

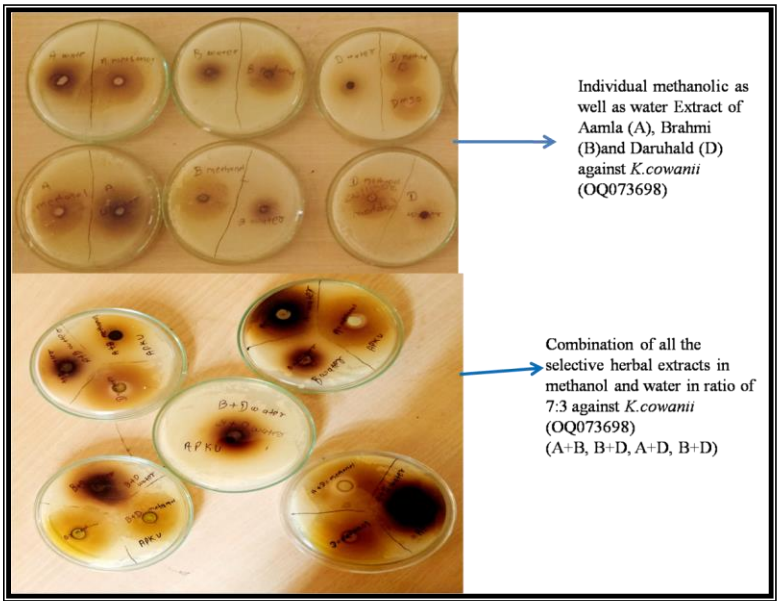


Figure 2: Antimicrobial activity of all extracts toward isolated uropathogen *Kosakonia cowanii* (OQ073698)

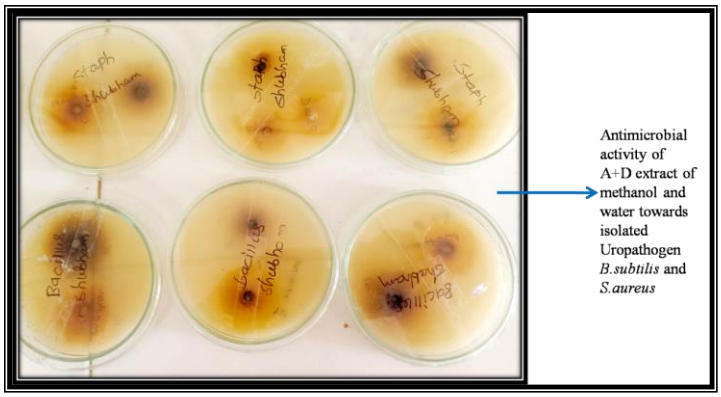


Figure 3: Antimicrobial activity of A+D extract toward *Bacillus subtilis* and *Staphylococcus aureus*

Successful extraction concerning bioactive compounds from row material depends on the solvent used in the extraction procedure. The recent study's main success is the use of methanolic and aqueous extracts from three plant materials that contained different active compounds with a synergistic effect in one preparation with particular concentration and combination of extracts [24]. The possible mechanism of action – methanolic extract for two plant materials-could contain different active compounds with a synergistic effect in one preparation, a phenolic compound known to alter microbial cell permeability, leading to change damage to the macromolecules inside the bacterial cell. Many enzymes responsible for either the bacterial reproduction, growth, or virulence factors might lose their activity in response to phenolic compounds [25-27].

It is reported that *Emblica officinalis* is diuretic. In Unani medicine, the dried fruits of amla are used to treat hemorrhage, diarrhea, and dysentery. It is adaptogenic, hepatoprotective, antitumor, hypocholesterolemic, antioxidant, and antiulcerogenic. A few fruits are also reported to be anti-inflammatory, analgesic, and antipyretic. Active constituents of amla have been found to be effective against *Staphylococcus aureus*, *Escherichia coli*, *Mycobacterium tuberculosis*, *Salmonella typhosa*, and *Candida albicans.* Similar findings were reported by Sabahat Saeed in 2010 [27].

Conclusion

As an alternative approach, the present study evaluates of herbal extracts that are traditionally used to treat urolithiasis. The uroprotective role of combination of two herbal extracts, *i.e., Emblica officinalis* and *Berberis aristata* has demonstrated the highest anti-UTI activity against the isolated uropathogens. These two herbal extracts are effective in treating UTI and can also serve as a replacement for the ampicillin antibiotics, as more uropathogens are resistant to ampicillin. The phytochemical test results suggested that the mixed extract of *Emblica officinalis* and *Berberis aristata* confirms the presence of flavonoides, alkaloids, saponins, and tannins that may interfere with the antimicrobial results against the isolated uropathogens.

ORCID

Amruta Prakashrao Kanakdande
https://orcid.org/0000-0003-4704-509X

References

[1] Medina, M., Castillo-Pino, E. An introduction to the epidemiology and burden of urinary tract infections. *Therapeutic Advances in Urology*, **2019**, 11, 3-7.
[2] Flores-Mireles, A.L., Walker, J.N., Caparon, M., Hultgren, S.J. Urinary tract infections: Epidemiology, mechanisms of infection and treatment options. *Nature Reviews Microbiology*, **2015**, 13(5), 269-284.
[3] Silverman, J.A., Schreiber IV, H.L., Hooton, T.M., Hultgren, S.J. From physiology to pharmacy: developments in the pathogenesis and treatment of recurrent urinary tract infections. *Current Urology Reports*, **2013**, 14(5), 448-456.
[4] Lewis, A.J., Richards, A.C., Mulvey, M.A. Invasion of host cells and tissues by uropathogenic bacteria. *Urinary Tract Infections: Molecular Pathogenesis and Clinical Management*, **2017**, 359-381.

[5] Khan, R., Islam, B., Akram, M., Shakil, S., Ahmad, A., Ali, S.M., Khan, A.U. Antimicrobial activity of five herbal extracts against multi drug resistant (MDR) strains of bacteria and fungus of clinical origin. *Molecules*, **2009**, 14(2), 586-597.
[6] Shaheen, G., Akram, M., Jabeen, F., Ali Shah, S.M., Munir, N., Daniyal, M., Sultana, S. Therapeutic potential of medicinal plants for the management of urinary tract infection: A systematic review. *Clinical and Experimental Pharmacology and Physiology*, **2019**, 46(7), 613-624.
[7] Gajdács, M., Dóczi, I., Ábrók, M., Lázár, A., Burián, K. Epidemiology of candiduria and Candida urinary tract infections in inpatients and outpatients: Results from a 10-year retrospective survey. *Central European Journal of Urology*, **2019**, 72(2), 209.
[8] Poloni, J.A.T., Rotta, L.N. Urine sediment findings and the immune response to pathologies in fungal urinary tract infections caused by *Candida spp. Journal of Fungi*, **2020**, 6(4), 245.
[9] Wawrysiuk, S., Naber, K., Rechberger, T., Miotla, P. Prevention and treatment of uncomplicated lower urinary tract infections in the era of increasing antimicrobial resistance—non-antibiotic approaches: A systemic review. *Archives of Gynecology and Obstetrics*, **2019**, 300(4), 821-828.
[10] Chardavoyne, P.C., Kasmire, K.E. Appropriateness of antibiotic prescriptions for urinary tract infections. *Western Journal of Emergency Medicine*, **2020**, 21(3), 633.
[11] Savoia, D. Plant-derived antimicrobial compounds: alternatives to antibiotics. *Future Microbiology*, **2012**, 7(8), 979-990.
[12] Ekor, M. The growing use of herbal medicines: issues relating to adverse reactions and challenges in monitoring safety. *Frontiers in Pharmacology*, **2014**, 4, 177.
[13] Howard, J. Substance Abuse Treatment for Persons with Child Abuse and Neglect Issues. Treatment Improvement Protocol (TIP) Series 36. *Substance Abuse and Mental Health Services Administration*, **2000**.
[14] World Health Organization (WHO). Our planet, our health.Report of WHO Commission on Health and Environment. Geneva: World Health Organization. **1992**.
[15] Sung, B., Chung, H.S., Kim, M., Kang, Y.J., Kim, D.H., Hwang, S.Y., Kim, N.D. Cytotoxic effects of solvent-extracted active components of Salvia miltiorrhiza Bunge on human cancer cell lines. *Experimental and Therapeutic Medicine*, **2015**, 9(4), 1421-1428.
[16] Naylor, N.R., Pouwels, K.B., Hope, R., Green, N., Henderson, K.L., Knight, G.M., Deeny, S.R. The health and cost burden of antibiotic resistant and susceptible Escherichia coli bacteraemia in the English hospital setting: A national retrospective cohort study. *PLoS One*, **2019**, 14(9), 221944.
[17] Snyder, M.D., Priestley, M.A., Weiss, M., Hoegg, C.L., Plachter, N., Ardire, S., Thompson, A. Preventing catheter-associated urinary tract infections in the pediatric intensive care unit. *Critical Care Nurse*, **2020**, 40(1), 12-17.
[18] Kanakdande, A.P., Jadhav, P.B. Anti-urinary tract infection activity of selected herbal extract towards isolated *Kosakonia cowanii* (OQ 073698). *Cellular, Molecular and Biomedical Reports*, **2023**, 3(2), 114-121.
[19] Numan, R.S. The use of fungi and biotechnology in treating environmental pollution. *South Asian Research Journal of Biology and Applied Biosciences*, **2026**, 8(2), 93-100.
[20] Kadhum, H.A., Hasan, T.H. The study of bacillus subtils antimicrobial activity on some of the pathological isolates. *International Journal of Drug Delivery Technology*, **2019**, 9(2), 193-196.
[21] Lew, A.L.Y., Mazlan, N., Anua, S.M. Screening of pathogenic bacteria from sea cucumber acaudina molpadioides in pulau langkawi, kedah. *Malaysian Journal of Medicine & Health Sciences*, **2020**, 16(3).
[22] Khameneh, B., Iranshahy, M., Soheili, V., Fazly Bazzaz, B.S. Review on plant antimicrobials: A mechanistic viewpoint. *Antimicrobial Resistance & Infection Control*, **2019**, 8(1), 118.
[23] Cai, T. Recurrent uncomplicated urinary tract infections: Definitions and risk factors. *GMS Infectious Diseases*, **2021**, 9, 3.
[24] La Storia, A., Ercolini, D., Marinello, F., Di Pasqua, R., Villani, F., Mauriello, G. Atomic force microscopy analysis shows surface structure changes in carvacrol-treated bacterial cells. *Research in Microbiology*, **2011**, 162(2), 164-172.
[25] Takó, M., Kerekes, E.B., Zambrano, C., Kotogán, A., Papp, T., Krisch, J., Vágvölgyi, C. Plant phenolics and phenolic-enriched extracts as antimicrobial agents against food-contaminating microorganisms. *Antioxidants*, **2020**, 9(2), 165.
[26] Hashim, S.T., Hamza, I.S., Hassan, M.A. Identification of quantative chemical compounds of ethanolic extracts of Quercus infectoria and studies its inhibitory effect in some bacteria. *Indian Journal of Research*, **2013**, 2(8), 125-128.
[27] Saeed, S. Herbal remedies for urinary tract infection. *International Journal of Biology and Biotechnology*, **2010**, 7(4), 347-352.

Copyright © 2026 by SPC (Sami Publishing Company) + is an open access article distributed under the Creative Commons Attribution License (CC BY), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.